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Roll No

IT-713-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Automata and Compiler Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All question carry equal marks.

1. a) Write the steps for the efficient construction of LALR parsing table. Explain with an example.
b) Construct CLR parser for the grammar $S \rightarrow L=R$, $S \rightarrow R$, $L \rightarrow *R$, $R \rightarrow L$
2. a) Draw a block diagram of phases of a compiler and indicate the main functions of each phase.
b) Explain bootstrapping a compiler with suitable diagrams.
3. a) Test whether the grammar is LL (1) or not, and construct a predictive parsing table for following grammar:
 $S \rightarrow iEtSS_1/a$, $S_1 \rightarrow eS / \epsilon$, $E \rightarrow b$
4. What are the principle sources of optimization? Give the classification of code optimization.

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5. a) What is Syntax directed translation? How it is different from translation schemes? Explain with an example.
b) Translate the given expression into Quadruples, triples and indirect triples:
$$(a + b) * (c + d) + (a * b / c) * b + 60$$
6. a) What is a three address code? Translate the following expression into:
i) Quadruples
ii) Triples
iii) Indirect triples
$$- (a + b) * (c + d) + (a + b + c)$$

b) Write a short note on Run Time Memory Management.
7. a) Explain in detail about Stack allocation scheme.
b) Explain any two storage allocation strategies with pros and cons.
8. Write short notes on the following (any two):
a) Peephole optimization
b) DAG
c) Register allocation and code selection
